

ngspice-46 vs. Spectre — Feature Gap Analysis

Ngspice + OpenVAF Enhancements project

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A capability comparison of the ngspice-46 build in this repository against a Spectre-class commercial simulator, to show where the open-source tool already matches the commercial one and where the gaps are. Grounded in a direct read of the ngspice-46/src/ tree (analyses, solver, devices, RF, convergence, parallelism), not marketing feature lists.

Legend: ✓ present / on par · ⚠ partial, experimental, or via a workaround · ✗ absent. "Spectre" stands in for the commercial reference (Spectre / SpectreRF / Spectre X / RelXpert feature set).

The one column that matters is ngspice — Spectre has essentially everything, so its column is a baseline of ✓. The point of the table is where ngspice's mark is ⚠ or ✗.

Context: the Verilog-A / OSDI device side is not a gap — thanks to the OpenVAF-reloaded compiler in this repository it is on par with (often ahead of) commercial Verilog-A support. The gaps below are all on the simulator/analysis side.

Core numerics

Category	Feature	ngspice	Spectre
Core solver	Sparse LU (KLU) direct solver	✓ ¹	✓
Core solver	Legacy Sparse1.3 fallback	✓	✓
Core solver	Parallel / partitioned / GPU linear solve	✗	✓
Core solver	Matrix reordering + scaling beyond KLU defaults	⚠	✓
Integration	Trapezoidal + variable-order Gear	✓	✓
Integration	Advanced LTE-based step/order control	⚠	✓
Convergence	gmin stepping + source stepping homotopy	✓	✓
Convergence	Pseudo-transient / dynamic-gmin continuation	⚠	✓
Convergence	Damped / trust-region (globalized) Newton	⚠	✓
Convergence	Coordinated accuracy presets (errpreset)	✓	✓

The two convergence ⚠ rows: ngspice's DC path is not naked — it has per-device junction limiting (30 device families), an optional `nodedamping` step clamp, and a multi-stage homotopy cascade (`dynamic_gmin` → `new_gmin` → `spice3_gmin` → source stepping). What it lacks vs. commercial tools is a principled globalized Newton (residual-based trust-region / Armijo) and a classic pseudo-transient ($\dot{X}$ -embedded) continuation. [Enhancement-111](#) adds the former: `.option linesearch`, an Armijo backtracking line search on a new KCL-residual merit $\|F\|=\|G\cdot x-b\|$ (result-neutral, off by default) — see the [implementation write-up](#). [Enhancement-112](#) then extended it to the KLU solver — it originally ran only under Sparse 1.3 and segfaulted under `.option klu` — so the line search now works under both linear solvers, with a residual-merit sequence numerically identical between them.

¹ KLU is compiled in (SuiteSparse, statically linked) but is not the default in this build — the default direct solver is Sparse 1.3; KLU is selected with `.option klu`. The two agree on DC / AC / transient, and — since [Enhancement-113](#) — on noise and single-ended pole-zero as well (the KLU adjoint solve was doing a non-transposed solve, silently wrong

on asymmetric matrices; now fixed), and — since [Enhancement-114](#) — on DC/AC sensitivity (`.sens`), and — since [Enhancement-115](#) — on distortion (`.disto`) too. The only analysis still Sparse-only under KLU is balanced-output pole-zero; KLU is also less robust on stiff transient edges. Full behavior, defaults, and a solver-by-solver sweep of the example suite: [KLU vs. Sparse 1.3 solver notes](#).

Standard analyses (analog)

Category	Feature	ngspice	Spectre
Analyses	DC operating point / DC sweep	✓	✓
Analyses	Transient (<code>.tran</code>)	✓	✓
Analyses	AC small-signal (<code>.ac</code>)	✓	✓
Analyses	Noise, small-signal (<code>.noise</code> , <code>onoise/inoise</code> + integrated)	✓	✓
Analyses	Pole-zero (<code>.pz</code>)	✓	✓
Analyses	Transfer function (<code>.tf</code>)	✓	✓
Analyses	DC sensitivity (<code>.sens</code>)	✓	✓
Analyses	Distortion (<code>.disto</code>)	✓	✓
Analyses	Measurement / post-processing (<code>.meas</code>)	✓	✓

RF / periodic steady-state suite

Category	Feature	ngspice	Spectre
RF	S-parameter analysis + noise figure (<code>.sp</code>)	✓	✓
RF	Touchstone (<code>.sNp</code>) import / export	✓	✓
RF	Periodic steady state (PSS)	✓ ¹	✓
RF	Harmonic Balance (HB)	✗	✓
RF	Periodic / phase noise (Pnoise)	⚠ ³	✓
RF	Periodic AC (PAC, conversion gain)	✓ ²	✓
RF	Periodic transfer function (PXF)	✓ ⁴	✓
RF	Periodic S-parameters (PSP)	✗	✓
RF	Quasi-periodic / multi-tone (QPSS / QPAC)	✗	✓
RF	Envelope following	✗	✓

¹ PSS was ⚠(experimental `--enable-pss` flag, so `.pss` was unimplemented in shipped builds, and ~230 lines of shooting-loop trace per run). Since [Enhancement-117](#) it is built by default, quiet (trace behind `set ngdebug`), and verified against the analytic AC response; [Enhancement-118](#) then made it run under both linear solvers (KLU had hung on a timestep explosion from reused refactor pivots — now a full re-factor is forced each PSS step under KLU). It is still a brute-force shooting method and remains the foundation for the periodic small-signal analyses below. HB exists only as a `WITH_HB` stub that returns “unsupported”.

² PAC is ✓ (complete periodic-AC analysis). The full chain is built and verified: [Enhancement-119](#) retains the periodic operating point, [Enhancement-120](#) extracts the periodic Jacobian harmonics G_k, C_k , [Enhancement-121](#) assembles and solves the $(2M+1)N$ harmonic conversion matrix $H_{\{nm\}} = G_{\{n-m\}} + j\omega_m C_{\{n-m\}}$, [Enhancement-122](#) wraps it in a user-facing `.pac` command (runs PSS, sweeps the input frequency, writes a complex plot), and [Enhancement-123](#) finishes it with a netlist-referenced small-signal source stimulus (a true transfer / conversion gain) and multi-sideband output — every conversion sideband $f_{in} + k \cdot f_0$ as its own named vector (`<node>_usb<k>/<node>_lsb<k>`). Verified on the RC: unit-current sideband-0 reproduces the AC driving-point impedance and, with `V1 AC 1`, sideband-0 reproduces the low-pass transfer $1/\sqrt{1+(2\pi fRC)^2}$ across the band while the conversion sidebands sit at floating-point zero (a linear circuit does not mix). The one scale caveat: the conversion matrix is solved densely (capped for modest circuits) on the brute-force shooting PSS. The same solve is the substrate for pnoise and PXF.

³ Pnoise is ⚠(working, stationary-source first cut). [Enhancement-124](#) adds a `.pnoise` command that folds every device’s noise through the adjoint of the conversion matrix ($H^T \Psi = e_{\{out,0\}}$): it loads the sideband- k transfer into `CKTrhs/CKTirhs` and calls the existing device noise routines (`NevalSrc`, `OSDI load_noise`) once per sideband, accumulating $\Sigma_k S \cdot |\Delta\Psi_k|^2$ — so it reuses every device noise model and needs no per-device code. Verified on the RC: because the circuit is linear the conversion matrix is block-diagonal, only sideband 0 contributes, and pnoise reduces exactly to `.noise` ($4kTR/(1+(2\pi fRC)^2)$, matching the `.noise` reference to every printed digit). [Enhancement-126](#) then adds a cyclostationary mode (`.pnoise ... cyclo`): it evaluates each device’s noise PSD at every PSS sample’s bias and folds it through the time-domain adjoint transfer, averaging over the period, so a pumped device’s bias-dependent noise

(a diode's $2qI(t)$, a resistor's flicker $\propto |I(t)|^2$) is captured correctly. It reduces exactly to the stationary case (and hence .noise) for a bias-independent source by Parseval, and on a flicker resistor carrying a known periodic current it gives $\text{noise} \cdot f = RI^2 \cdot KF \cdot \langle I^2 \rangle$ using the period-average $\langle I^2 \rangle$ (matched to five digits). It stays  only because it does not yet emit a dedicated phase-noise (jitter) spectrum — a refinement of the same cyclostationary fold.

*4 PXF is  (complete). [Enhancement-125](#) adds a .pxf command — the adjoint of PAC. It solves $H^T \Psi = e_{\{\text{out}, 0\}}$ per frequency and dots each sideband block of Ψ with the netlist AC-source pattern B0 to get the input→output transfer at each sideband ($\times f, \times f_{\text{usb}} < k >, \times f_{\text{lsb}} < k >$). By the identity $(H^{-1}B)_{\text{out}} = (H^{-T}e_{\text{out}})^T B$ the sideband-0 transfer is bit-identical to the PAC response at the output — a reciprocity cross-check that pins the adjoint solve — verified on the RC to equal the analytic low-pass transfer, with the conversion sidebands at floating-point zero. The same dense-solve scale caveat as PAC applies. This completes the PSS → PAC → Pnoise → PXF periodic small-signal suite.

Performance & scale

Category	Feature	ngspice	Spectre
Performance	Multithreaded device-model evaluation (OpenMP)		
Performance	Element bypass / latency exploitation		
Performance	Fast-SPICE hierarchical / isomorphic engine		
Performance	Distributed (MPI) / cloud partitioning		
Performance	Handles 10^6 – 10^8 -node post-layout netlists		

Statistical / variability / yield

Category	Feature	ngspice	Spectre
Statistical	Monte Carlo		
Statistical	Native process/mismatch modeling + correlations		
Statistical	Low-discrepancy sampling (Sobol / Latin-hypercube)		
Statistical	High-sigma methods (importance sampling, worst-case distance)		
Statistical	Corner + MC + yield estimation flow		

Monte Carlo is  works, but script-driven (`alter + sgauss`, with the deterministic-seed RNG from [Enhancement-10/66](#)) rather than a native statistical block with sampling controls.

Reliability / aging

Category	Feature	ngspice	Spectre
Reliability	Device aging (HCI / NBTI / TDDB)		
Reliability	Stress → degrade → re-simulate (fresh/aged) flow		
Reliability	Electromigration + IR-drop (EMIR)		

Post-layout / parasitics

Category	Feature	ngspice	Spectre
Post-layout	Flat parasitic (RC) netlist simulation		
Post-layout	RC reduction / model-order reduction		
Post-layout	n-port (S/Y/Z) extracted-block import		

Behavioral / mixed-signal / AMS

Category	Feature	ngspice	Spectre
Modeling	Verilog-A (analog, LRM Annex C) via OSDI	✓	✓
Modeling	XSPICE code models + event-driven digital	✓	✓
Modeling	Verilog-AMS mixed-signal (connect modules)	×	✓
Modeling	Real-number modeling (wreal / RNM)	×	✓
Modeling	VHDL-AMS	×	✓

Interfaces & infrastructure

Category	Feature	ngspice	Spectre
Interface	Shared-library / programmatic API	✓	✓
Interface	Python bindings	✓	✓
Infrastructure	Built-in optimizer	×	✓
Infrastructure	Checkpoint / restart of long runs	⚠	✓

Where to invest (given the Verilog-A/OSDI side is done)

The realistic, winnable identity for this project is the open-source, OSDI/Verilog-A-native simulator with a real RF-noise and statistical story — not out-engineering 25 years of commercial fast-SPICE/parallel work. Ranked by leverage on the existing strength $\times$ differentiation $\times$ tractability:

1. RF periodic small-signal suite — PAC $\rightarrow$ Pnoise $\rightarrow$ PXF, on the hardened PSS. The PSS foundation is now shipped and verified under both linear solvers ([Enhancement-117](#), [Enhancement-118](#)), and [Enhancement-119](#) retains the periodic operating point (voltages + device states per sample) and [Enhancement-120](#) turns it into the periodic Jacobian harmonics G_k , C_k , and [Enhancement-121](#) assembles those into the $(2M+1)N$ harmonic conversion matrix and solves it, [Enhancement-122](#) wraps that in a working `.pac` command, and [Enhancement-123](#) finishes it with a netlist-referenced source stimulus and multi-sideband conversion-gain output — a complete periodic-AC analysis, verified against the exact linear transfer and driving-point responses, [Enhancement-124](#) adds `.pnoise` — folding every device's noise through the conversion-matrix adjoint H^T (reusing the existing device noise routines), verified to reduce exactly to `.noise` in the linear limit — and [Enhancement-125](#) adds `.pxf`, the adjoint transfer function, whose sideband-0 result is bit-identical to the PAC response by reciprocity. The PSS $\rightarrow$ PAC $\rightarrow$ Pnoise $\rightarrow$ PXF periodic small-signal suite is now complete — genuinely novel in open source — and [Enhancement-126](#) adds cyclostationary noise (per-sample bias, time-domain-transfer fold) so pumped devices' bias-dependent noise is handled correctly. The remaining RF work is the further refinements (a dedicated phase-noise/jitter spectrum, a from-scratch Harmonic Balance engine, quasi-periodic multi-tone) rather than the core analyses.
2. Convergence robustness — coordinated accuracy presets (`errpreset`) landed in [Enhancement-110](#); the remaining piece is pseudo-transient / dynamic-gmin homotopy. Unglamorous but makes every analysis usable on real circuits; self-contained in the ngspice core.
3. High-sigma statistical sampling — high industrial value (yield / SRAM), moderate difficulty, leans on the deterministic-seed RNG already in place.

Explicitly lower priority: fast-SPICE parallelism and aging/EMIR — enormous efforts that don't leverage what makes this project distinctive.